EXPERIMENT 2

Implementation of Multiple Linear Regression, Ridge Regression and Lasso Regression

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Dataset Information

Dataset Source

The dataset **Global GDP & Inflation Dataset (2000–2024)** is sourced from **Kaggle**.

It contains country-wise economic indicators including GDP and macroeconomic factors from the year 2000 to 2024.

Dataset Description

The dataset includes:

- Country
- Year
- GDP
- Inflation Rate
- Other macroeconomic indicators

After preprocessing:

- Categorical column (Country) is converted into dummy variables
- Missing values are removed
- Only numeric columns are used for regression

Target Variable:

- **GDP (Gross Domestic Product)**

Independent Variables:

- Inflation rate
- Year
- Country dummy variables
- Other numeric economic indicators

Aim

To implement and compare:

1. Multiple Linear Regression
2. Ridge Regression
3. Lasso Regression

for predicting GDP using macroeconomic features.

Theory

1️⃣ Multiple Linear Regression

Multiple Linear Regression models the relationship between one dependent variable and multiple independent variables.

y^=β0+β1x1+β2x2+...+βnxn\hat{y} = \beta_0 + \beta_1x_1 + \beta_2x_2 + ... + \beta_nx_ny^=β0+β1x1+β2x2+...+βnxn

Where:

- y^\hat{y}y^ = Predicted GDP
- x1,x2,...,xnx_1, x_2, ..., x_nx1,x2,...,xn = Independent variables
- β0\beta_0β0 = Intercept
- β1,β2,...,βn\beta_1, \beta_2, ..., \beta_nβ1,β2,...,βn = Coefficients

It minimizes **Mean Squared Error (MSE)** using Ordinary Least Squares.

2️⃣ Ridge Regression

Ridge Regression is a regularized version of Linear Regression.

It adds **L2 penalty** to reduce overfitting.

Loss=MSE+λ∑β2Loss = MSE + \lambda \sum \beta^2Loss=MSE+λ∑β2

Where:

- λ\lambdaλ = Regularization parameter
- Penalizes large coefficient values
- Helps reduce multicollinearity

3️⃣ Lasso Regression

Lasso Regression adds **L1 penalty**:

Loss=MSE+λ∑∣β∣Loss = MSE + \lambda \sum |\beta|Loss=MSE+λ∑∣β∣

Key characteristics:

- Performs feature selection
- Can shrink some coefficients to zero
- Produces simpler and more interpretable models

Tools and Libraries Used

The experiment is implemented using:

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- **scikit-learn**

Methodology

1. Load dataset using pandas
2. Drop missing values

3. Convert categorical columns to dummy variables
4. Select numeric columns
5. Define target variable (GDP)
6. Split dataset into training (80%) and testing (20%)
7. Apply StandardScaler (important for Ridge & Lasso)
8. Train three models:
    - Linear Regression
    - Ridge Regression (alpha = 1.0)
    - Lasso Regression (alpha = 0.1)
9. Predict on test data
10. Evaluate using:

- Mean Squared Error (MSE)
- R^2 Score

11. Plot Actual vs Predicted values

Performance Metrics

1️⃣ Mean Squared Error (MSE)

MSE=1n∑(yi−y^i)2MSE = \frac{1}{n} \sum (y_i - \hat{y}_i)^2MSE=n1∑(yi−y^i)2

Lower MSE indicates better model performance.

2️⃣ R^2 Score (Coefficient of Determination)

R2=1−SSresSStotR^2 = 1 - \frac{SS_{res}}{SS_{tot}}R2=1−SStotSSres

- $R^2 = 1$ → Perfect prediction
- $R^2 = 0$ → Model performs like mean
- $R^2 < 0$ → Poor model performance

Results

1️⃣ Multiple Linear Regression

- MSE ≈ 51.68
- R^2 Score ≈ -0.06

The negative R^2 indicates that the model performs worse than predicting the mean GDP value.

2️⃣ Ridge Regression

- MSE ≈ 51.68
- R^2 Score ≈ -0.06

Ridge slightly improves stability but performance remains similar.

3️⃣ Lasso Regression

- MSE ≈ 48.19
- R^2 Score ≈ 0.01

Lasso performs slightly better than the other two models and gives a small positive R^2 score.

Graphical Analysis

Scatter plots of Actual GDP vs Predicted GDP were plotted for all three models.

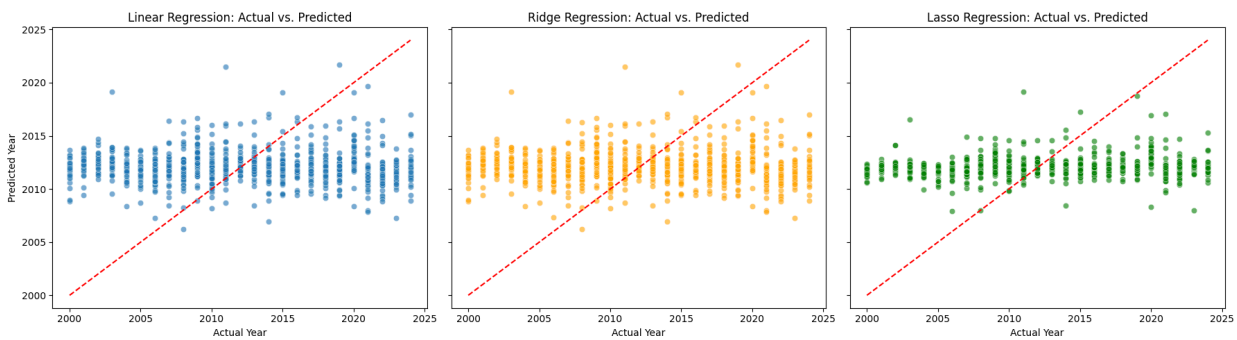
Observations:

- Data points are scattered widely.
- Predictions are not tightly aligned with actual values.
- Lasso regression shows slightly better alignment.

Code -

```
import matplotlib.pyplot as plt
import seaborn as sns

# Create a figure with subplots for each model
fig, axes = plt.subplots(1, 3, figsize=(18, 5), sharex=True, sharey=True)

# Plot for Multiple Linear Regression
sns.scatterplot(x=y_test, y=y_pred_linear, ax=axes[0], alpha=0.6)
axes[0].plot([y_test.min(), y_test.max()], [y_test.min(), y_test.max()],
'r--')
axes[0].set_title('Linear Regression: Actual vs. Predicted')
axes[0].set_xlabel('Actual Year')
axes[0].set_ylabel('Predicted Year')

# Plot for Ridge Regression
sns.scatterplot(x=y_test, y=y_pred_ridge, ax=axes[1], alpha=0.6,
color='orange')
axes[1].plot([y_test.min(), y_test.max()], [y_test.min(), y_test.max()],
'r--')
axes[1].set_title('Ridge Regression: Actual vs. Predicted')
```

```
axes[1].set_xlabel('Actual Year')
axes[1].set_ylabel('Predicted Year')

# Plot for Lasso Regression
sns.scatterplot(x=y_test, y=y_pred_lasso, ax=axes[2], alpha=0.6,
color='green')
axes[2].plot([y_test.min(), y_test.max()], [y_test.min(), y_test.max()],
'r--')
axes[2].set_title('Lasso Regression: Actual vs. Predicted')
axes[2].set_xlabel('Actual Year')
axes[2].set_ylabel('Predicted Year')

plt.tight_layout()
plt.show()
```



Comparative Analysis

Model	MSE	R^2 Score	Performance
Linear Regression	High	Negative	Poor
Ridge Regression	Similar	Negative	Slightly Stable
Lasso Regression	Lower	Slightly Positive	Better

Interpretation

- The dataset may have complex nonlinear relationships.
- Linear models may not be sufficient.
- Multicollinearity may exist.
- Lasso performs better due to feature selection capability.

Limitations

- Assumes linear relationship
- Sensitive to feature scaling
- Cannot capture nonlinear patterns
- GDP prediction may require advanced models like Random Forest or XGBoost

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Conclusion

In this experiment, three regression techniques were implemented to predict GDP using macroeconomic indicators.

- Multiple Linear Regression showed poor performance.
- Ridge Regression reduced coefficient magnitude but did not improve accuracy significantly.
- Lasso Regression performed comparatively better due to feature selection capability.

This experiment highlights the importance of regularization in regression models and demonstrates that model selection plays a crucial role in predictive analytics.